

Deepak B. Waman (M.Sc., Ph.D.)

Postdoctoral Researcher | Cloud Microphysics, Numerical Modeling | KIT, Germany

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Research Statement

As an atmospheric scientist, I investigate cloud and precipitation processes and aerosol–cloud–radiation interactions by combining high-resolution numerical modeling with observations from research aircraft, ground-based, and spaceborne instruments. My research focuses on understanding how ice nucleation and multiplication influence cloud glaciation, precipitation efficiency, and cloud radiative properties in convective systems, with the goal of improving the representation of these processes in atmospheric models. In my postdoctoral research, I integrate, compare, and evaluate cloud microphysical parameterizations within the ICON model by combining observations. I have also been exploring computationally efficient approaches for representing ice multiplication, including machine learning techniques. In parallel, I investigate how aerosols and environmental conditions influence lightning activity across different convective regimes using global observational datasets. My long-term research goal is to understand how cloud microphysical processes and environmental conditions govern thunderstorm electrification and lightning activity, particularly over the Indian monsoon region.

Research Interests

- Ice nucleation and multiplication, convective cloud dynamics
- Aerosol-cloud interactions and radiative effects
- High-resolution numerical modelling
- Aerosol-cloud-lightning on regional and global scale and thunderstorm electrification

Education

Oct. 2019 – Oct. 2023	<i>PhD</i> , Department of Earth and Environmental Sciences, Lund University, Sweden <i>Thesis</i> : Mechanisms for the Influence from Ice Nucleus Aerosols on Clouds and their Indirect Effects: Cloud Modelling <i>Supervisor</i> : Dr. Vaughan Phillips
Jul. 2017 – June 2019	<i>M.Sc.</i> , Atmospheric Sciences, Savitribai Phule Pune University, Pune, India 74.9%, CGPA: 9.0/10
June 2013 – June 2016	<i>B.Sc.</i> , Physics, Savitribai Phule Pune University, Pune, India 90.16%

Professional Experience

Dec. 2023 - Present	<i>Postdoctoral Researcher</i> , Institute of Meteorology and Climate Research Troposphere Research (IMKTRO), Karlsruhe Institute of Technology (KIT), Germany <i>Project</i> : WarmWorld (German Federal Ministry of Education and Research) Km-scale ICON model development for climate projections <i>Focus</i> : Improving mixed-phase and ice clouds representation in one- and two-moment cloud microphysics schemes of ICON for km-scale climate simulations.
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Supervisor: Prof. Corinna Hoose

Independent side projects: Influence of aerosols, cloud microphysics, and thermodynamic environments on global lightning activity | Data-driven approaches for representing ice multiplication in climate models

Teaching Experience

- Oct. 2025 - Feb. 2026 Tutor and Lecturer - Cloud Physics (Master's level), KIT, Germany
- Prepared, designed and delivered lectures on Parcel Theory, Cloud microphysics and storm Dynamics, and tutorials to 20 M. Sc. Students
 - Designed interactive quizzes (Mentimeter) on relevant topics
- Oct. 2024 – Feb. 2025 Tutor - Cloud Physics (Master's level), KIT, Germany
- Conducted problem solving sessions for Master's students (14 students)

Teaching interests

- Cloud Physics and Atmospheric Thermodynamics
- Atmospheric/Cloud Dynamics
- Numerical and Computational Techniques in Atmospheric Science
- Atmospheric Electricity and Thunderstorm Electrification
- Computational Lab (Fortran, Python) and Numerical Modelling (WRF, ICON)

Student Supervision

- Ann Henze **MSc thesis.** Data-driven approach to understand controls on lightning occurrence across global lightning hotspots (Main Supervisor)
- Xuan Zhang **MSc thesis.** Predicting ice multiplication using machine learning (KIT, 2026) (Main Supervisor)
- Antonis Michalopoulos **MSc thesis.** Impact of SIP on the dynamics of extratropical Cyclones (KIT, 2026) (Co-supervisor)
- Abdullah Nassar **Research assistant,** Influence of soluble aerosols on lightning activity over Europe (KIT, 2025-2026) (Main Supervisor)

Peer-Reviewed Publications

h-index	8		i10-index	5		citations	205
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A. First-author

a. Published

1. **Waman D.**, Meusel, J., Keshtgar, B., Wallentin, G., Barthlott, C., Patade, S., Shete, S., Prabhakaran, T., Fievet, R., Finney, D., Blyth, A., & Hoose, C., 2026. Impacts of Secondary Ice Production on the Microphysics and Dynamics of Deep Convective Clouds in Different Environments. Atmospheric Chemistry and Physics. <https://doi.org/10.5194/acp-26-11371-2026>.
2. **Waman, D.**, Jadav, A., Patade, S., Gautam, M., Deshmukh, A. and Phillips, V., 2025. Mechanisms for Indirect Effects from Ice Nucleating Particles on Continental Clouds and Radiation. Journal of the Atmospheric Sciences, 82(8), pp.1693-1720. <https://doi.org/10.1175/JAS-D-24-0143.1>.
3. **Waman, D.**, Deshmukh, A., Jadav, A., Patade, S., Gautam, M., Phillips, V., Bansemer, A. and Jakobsson, J., 2023. Effects from time dependence of ice nucleus activity for contrasting cloud types. Journal of the Atmospheric Sciences, 80(8), pp.2013-2039. <https://doi.org/10.1175/JAS-D-22-0187.1>.
4. **Waman, D.**, Patade, S., Jadav, A., Deshmukh, A., Gupta, A.K., Phillips, V.T., Bansemer, A. and DeMott, P.J., 2022. Dependencies of Four Mechanisms of Secondary Ice Production on Cloud- Top Temperature in a Continental Convective Storm. Journal of the Atmospheric Sciences, 79(12), pp.3375-3404. <https://doi.org/10.1175/JAS-D-21-0278.1>.

b. In Preparation

1. **Waman, D.**, 2026. Contrasting effects of insoluble aerosols on extreme lightning across continental hotspots. (to be submitted to Geophysical Research Letters)
2. **Waman D.**, Nassar, A., Hoose, C., 2026. Aerosols and ice fraction governs global lightning (to be submitted to Atmospheric Chemistry and Physics)

B. Co-author

1. Pant, C.S., **Waman, D.**, *et al.*, 2026. A modified stratiform cloud microphysics parameterization: evaluation using the Community Atmosphere Model version 6 single-column model. *Atmospheric Chemistry and Physics*, 26(10), pp.7407-7433. <https://doi.org/10.5194/acp-26-7407-2026>.
2. Jadav, A., **Waman, D.**, *et al.*, 2025. An improved convection parameterization with detailed aerosol–cloud microphysics for a global model. *Journal of the Atmospheric Sciences*, 82(1), pp.197-231, 2025. <https://doi.org/10.1175/JAS-D-23-0175.1>.
3. Patade, S., Kulkarni, G., Patade, S., **Waman, D.**, *et al.*, 2025. Importance of secondary ice production in mixed-phase monsoon clouds over the Indian subcontinent. *Atmospheric Research*, 315, p.107890. <https://doi.org/10.1016/j.atmosres.2024.107890>.
4. Gautam, M., **Waman, D.**, *et al.*, 2024. Fragmentation in Collisions of Snow with Graupel/Hail: New Formulation from Field Observations. *Journal of the Atmospheric Sciences*, 81(12), pp.2149-2164. <https://doi.org/10.1175/JAS-D-23-0122.1>.
5. Gupta, A.K., Deshmukh, A., **Waman, D.**, *et al.*, 2023. The microphysics of the warm-rain and ice crystal processes of precipitation in simulated continental convective storms. *Communications Earth & Environment*, 4(1), p.226. <https://doi.org/10.1038/s43247-023-00884-5>.
6. Patade, S., **Waman, D.**, *et al.*, 2022. The influence of multiple groups of biological ice nucleating particles on microphysical properties of mixed-phase clouds observed during MC3E. *Atmospheric Chemistry and Physics*, 22(18), pp.12055-12075. <https://doi.org/10.5194/acp-22-12055-2022>.
7. Jakobsson, J.K., **Waman, D.**, *et al.*, 2022. Time dependence of heterogeneous ice nucleation by ambient aerosols: laboratory observations and a formulation for models. *Atmospheric Chemistry and Physics*, 22(10), pp.6717-6748. <https://doi.org/10.5194/acp-22-6717-2022>.
8. Deshmukh, A., Phillips, V.T., Bansemer, A., Patade, S. and **Waman, D.**, 2022. New empirical formulation for the sublimational breakup of graupel and dendritic snow. *Journal of the Atmospheric Sciences*, 79(1), pp.317-336. <https://doi.org/10.1175/JAS-D-20-0275.1>.

Awards and Achievements

2017 IIT-JAM (Physics): All India Rank 1775

2016 Best Student Award: Sangamner College, Sangamner (Affiliated to Savitribai Phule Pune University, Pune)

Professional Service

Peer-reviewer for:

- Atmospheric Chemistry and Physics (EGU/Copernicus)
- Journal of Geophysical Research – Atmospheres (AGU)
- Springer Nature journals
- Advances in Atmospheric Sciences (Science Press and Springer)

Selected Conferences and Workshops

Apr. 2026 Pre-EGU Ice-Cloud Workshop, University of Vienna, Austria

May 2026 European Geoscience Union (EGU), Vienna, Austria

Poster: Regime-dependent Impacts of CCN and Cloud Glaciation on Global Lightning Activity, DOI: 10.5194/egusphere-egu26-3745.

- Apr. 2024 Pre-EGU Ice-Cloud Workshop, University of Vienna, Austria
European Geoscience Union (EGU), Vienna, Austria
Talk: Importance of Age of Convective Clouds for Explosive Ice Crystal Number Growth via Secondary Ice Production, DOI: 10.5194/egusphere-egu24-2859.
Poster: Sensitivity of Microphysical Properties of Mixed-Phase Clouds on Model Resolution and Microphysics Scheme in ICON, DOI: 10.5194/egusphere-egu24-11022.
- June 2024 ICON Numerical Model Training Course, Deutscher Wetterdienst (DWD), Offenbach, Germany
- Apr. 2023 European Geoscience Union (EGU), Vienna, Austria
Poster: Singular versus Time-dependent Approach of Heterogeneous Ice Nucleation and Secondary Ice Production in Contrasting Clouds, DOI: 10.5194/egusphere-egu23-1834.
- Dec. 2022 American Geophysical Union (AGU), Chicago, USA
Oral: Primary and Secondary Ice Formation in Various Clouds: A numerical Study
- Aug 2022 American Meteorological Society, Collective Madison Meeting, Wisconsin, USA
Poster: Role of Four SIP Mechanisms in Ice Enhancement in Continental Convective Clouds: A Numerical Study
- Aug. 2021 International Commission for Clouds and Precipitation, Indian Institute of Tropical Meteorology, Pune, India
Poster: Dependency of Ice Enhancement Ratio on Cloud Top Temperature in Continental Deep Convective Clouds

Referees

Prof. Corinna Hoose

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